

Itnavideo: All Video Types Deep Reference

Complete Technical Blueprint covering every video template in Itnavideo: Purpose, Codebase Wiring, Architecture, AI Pipelines, Remotion Compositions, Billing Math, and Production Failure Recovery.

DOCUMENT CATEGORY

Video Types Engine & Template Registry

CORE TECHNOLOGIES

Remotion, React, Groq Whisper, AWS Lambda, S3

SCOPE

All 11 Video Formats & Creator Pipelines

TARGET AUDIENCE

Founders, Full-Stack Engineers & Video Architects

Master Video Types Index

#	Video Type Title	Aspect Ratio	Max Duration	Credit Formula	Pipeline Type
1	YouTube Subtitle Generator	16:9 Landscape (1920×1080)	15 Minutes (900s)	1 Credit / 2 Min	Fast-Path (Groq + Remotion)
2	Auto Caption Generator	9:16 Vertical (1080×1920)	90 Seconds	1 Credit Flat	Fast-Path (Groq + Remotion)
3	Compare Explainer Video	9:16 Vertical Split-Screen	60 Seconds	1 Credit Flat	Asset Matcher + SFX Manifest
4	Long Video Promo	9:16 Vertical Teaser	60 Seconds	1 Credit Flat	Instant Overlay (Skip Transcription)
5	Whiteboard Video	9:16 Vertical Auto-Draw	60 Seconds	1 Credit Flat	Speech-Synced Whiteboard Engine
6	Typography Video	9:16 Vertical Kinetic Text	60 Seconds	1 Credit Flat	Dynamic Keyword Shockwave
7	Long Video Clips	9:16 Podcast/Repurposing	60 Seconds	1 Credit Flat	Clip Extract + Highlight Subtitles
8	Faceless Video / AI Video Generator	16:9 Landscape Documentary	5 - 20 Minutes	Tiered Credits	AI Scene Planner + B-Roll + Canva
9	Image To Video AI	16:9 Landscape Storyteller	3 Minutes	Tiered Credits	Ken Burns Motion + Voice Sync
10	Creator Background Replace	9:16 Vertical Studio	60 Seconds	1 Credit Flat	Segmentation + Layer Stacking
11	AI Audio Cleaner	Audio Voice Stems	15 Minutes	Free / 1 Credit	Echo & Noise Filter + Pause Cutter

1. YouTube Subtitle Generator

16:9 LANDSCAPE

FAST-PATH ENGINE

1080P FULL HD

1.1 Identity & Specs

- **Dashboard Mode:** `mode: "youtubeSubtitleGenerator"`
- **AI Registry Template:** `YOUTUBE_SUBTITLE_GENERATOR`
- **Remotion Composition ID:** `AUTO-CAPTION-GENERATOR-LANDSCAPE`
- **Resolution & Frame Rate:** 1920 × 1080 (16:9 Landscape), 30 FPS
- **Duration Cap:** Strictly up to **15 minutes (900 seconds)**.
- **Pricing Formula:** `calculateYouTubeSubtitleCreditUnits(durationSeconds)` = **1 credit per 2 minutes** (120s blocks).
Max 8 credits for 15 mins.

1.2 Purpose & Creator Problem Solved

Long-form YouTube creators spend 3 to 6 hours typing and timing subtitles in Adobe Premiere or hiring freelancers on Fiverr. YouTube Subtitle Generator automates this completely in ~2 minutes with 5 high-retention creator styles.

1.3 Codebase Wiring

Layer	File Path	Responsibility
Frontend Dashboard	<code>app/dashboard/page.tsx</code>	Registers in <code>videoTypeCards</code> (long category), manages 15-min duration validation, and renders pricing badges.
Planner Registry	<code>services/ai/reelPlanner.ts</code>	Maps <code>YOUTUBE_SUBTITLE_GENERATOR</code> with <code>skipPlanner: true</code> and <code>needsCaptionStylePicker: true</code> .
Fast-Path API Route	<code>app/api/reels/jobs/route.ts</code>	Bypasses LLM planner, sends audio straight to Groq Whisper, and invokes <code>renderMediaOnLambda()</code> .
Remotion Composition	<code>remotion/templates/AUTO_CAPTION_GENERATOR/</code>	Renders video track and overlays word-synced subtitles in 16:9 bounds.
Client Preview	<code>components/preview/VideoTypePreviewRenderer.tsx</code>	Renders live 16:9 Remotion Player without charging credits.

1.4 The 5 Curated Creator Styles

- **Minimal Clean:** Inter Sans, translucent dark pill background (Ali Abdaal / Veritasium).
- **Cinematic Docu:** Montserrat Bold, off-white text with gold keywords (Vox / Johnny Harris).
- **Studio Podcast:** Poppins Bold, matte black box for 100% legibility on busy video backdrops (Diary of a CEO).
- **Bold Creator:** Montserrat ExtraBold, black outline with yellow pop animations (MrBeast / MKBHD).
- **Netflix Classic:** Roboto Condensed, broadcast cinema yellow with drop shadow.

1.5 Common Errors & Technical Fixes

Error: Video Aspect Ratio Squishing / Black Bars

Root Cause: User uploads a vertical 9:16 or square 1:1 video into YouTube Subtitle Generator.

Fix: In `remotion/templates/AUTO_CAPTION_GENERATOR/template.tsx`, the `<Video />` element uses `object-fit:`

`contain` with a blurred mirrored background behind non-16:9 video so no content is stretched.

2. Auto Caption Generator (Reels / Shorts / TikTok)

9:16 VERTICAL

FAST-PATH ENGINE

2.1 Identity & Specs

- **Dashboard Mode:** `mode: "autoCaption"`
- **AI Registry Template:** `AUTO_CAPTION_GENERATOR`
- **Remotion Composition ID:** `AUTO-CAPTION-GENERATOR`
- **Resolution & Frame Rate:** 1080 × 1920 (9:16 Vertical), 30 FPS
- **Duration Cap:** Up to **90 seconds** (optimal for Instagram Reels, YouTube Shorts, TikTok).
- **Pricing:** 1 credit flat.

2.2 Purpose & Creator Problem Solved

Over 75% of social media users watch Reels and TikTok videos on mute. Without animated subtitles, viewers scroll away within the first 2 seconds. Auto Caption Generator adds kinetic, word-by-word bouncing captions with active word highlighting.

2.3 Codebase Wiring

- `remotion/components/MotionCaptionRenderer.tsx` : Calculates the exact millisecond frame window for each word using `useCurrentFrame()` and applies pop scale transformations.
- `remotion/components/library/UniversalCaptionLayer.tsx` : Handles safe-area padding to avoid Instagram and TikTok UI buttons (like, comment, share icons).

2.4 Common Errors & Technical Fixes

Error: Subtitle words overlapping screen buttons on TikTok/Instagram

Root Cause: Subtitle position is too close to the bottom edge.

Fix: Safe margins of `bottom: 240px` are enforced in `UniversalCaptionLayer.tsx` to keep text completely visible above user interface controls.

3. Compare Explainer Video

9:16 VERTICAL

AI SCENE MATCHER

3.1 Identity & Specs

- **Dashboard Mode:** `mode: "compare"`
- **AI Registry Template:** `comparisonImages`
- **Remotion Composition ID:** `comparisonImages` (mapped to `CompareExplainer`)
- **Resolution & Frame Rate:** 1080 × 1920 (9:16 Vertical), 30 FPS
- **Duration Cap:** 60 Seconds
- **Required User Inputs:** 1 Audio voiceover + 2 Images (Left Item vs Right Item) + Titles (e.g. "iPhone 16" vs "S24 Ultra").

3.2 Purpose & Creator Problem Solved

Comparison videos ("Product A vs Product B", "Gym vs Home Workout", "Investing vs Real Estate") drive massive viral engagement on Reels. This template creates a split-screen layout with automated VS badges, sound effects, and voiceover synchronization.

3.3 Codebase Wiring

Layer	File Path	Responsibility
Composition Template	<code>remotion/templates/COMPARE_EXPLAINER/template.tsx</code>	Renders split layout: Left image, Right image, center VS emblem, progress bar.
Sound Effects Manifest	<code>remotion/templates/COMPARE_EXPLAINER/sfxManifest.ts</code>	Triggers punchy sound effects (whoosh, hit, gavel, buzzer) at key speech transitions.
Feature Flags	<code>services/ai/reelPlanner.ts</code>	<code>needsImages: true</code> , <code>needsCompareFields: true</code> .

3.4 Common Errors & Technical Fixes

Error: Images distorted or unevenly cropped in split screen

Root Cause: Left and right images have drastically different aspect ratios.

Fix: The template wraps each side in a responsive flexbox container with `overflow: hidden` and CSS `object-fit: cover` with subtle Ken Burns auto-pan so both sides look cinematic.

4. Long Video Promo

9:16 VERTICAL

ULTRA-FAST RENDER

4.1 Identity & Specs

- **Dashboard Mode:** `mode: "LongVideoPromo"`
- **Remotion Composition ID:** `LONG-VIDEO-PROMO`
- **Inputs:** 1 Video clip (10–60s) + 1 YouTube Thumbnail Image + Video Title.
- **Execution Type:** `skipTranscription: true`, `skipPlanner: true` (zero AI wait time).

4.2 Purpose & Creator Problem Solved

YouTube creators need to promote their new 20-minute video on Instagram Reels or YouTube Shorts. This template overlays the YouTube thumbnail card at the top or bottom of a video teaser clip with an animated "Watch Full Video on YouTube" call-to-action.

5. Whiteboard Video (Auto-Draw)

9:16 VERTICAL

EDUCATIONAL EXPLAINER

5.1 Identity & Specs

- **Dashboard Mode:** `mode: "whiteboardVideo"`
- **Remotion Composition ID:** `WHITEBOARD-VIDEO`
- **Inputs:** Spoken audio or video explainer.
- **Resolution:** 1080 × 1920 (9:16 Vertical), 30 FPS.

5.2 How the Auto-Draw Whiteboard Engine Works

The system analyzes the speech transcript using NLP keyword extraction, creates bullet points and diagrams, and simulates hand-drawn SVG sketch reveals synchronized to speech pacing using Remotion's `interpolate()` `stroke-dashoffset` animations.

6. Typography Video

9:16 VERTICAL KINETIC SHOCKWAVE

6.1 Identity & Specs

- **Dashboard Mode:** mode: "typographyVideo"
- **Remotion Composition ID:** TYPOGRAPHY-VIDEO
- **Feature Flag:** needsTypographyStylePicker: true
- **Inputs:** Spoken voiceover or talking video.

6.2 Purpose & Visual Treatment

Ideal for quotes, motivational speeches, statistics, and high-impact announcements. Instead of regular small subtitle bars, large bold typography fills the canvas, with words snapping into place with 3D camera shakes, zooms, and neon glows.

7. Long Video Clips (Podcast Repurposing)

9:16 VERTICAL CONTENT REPURPOSING

7.1 Identity & Specs

- **Dashboard Mode:** mode: "longVideoClips"
- **Remotion Composition ID:** LONG-VIDEO-CLIPS
- **Inputs:** Long video file (podcast, interview, or lecture).
- **Output:** 60-second high-energy vertical clip with creator captions.

7.2 How It Works

The audio is transcribed to locate peak energy moments (highest words-per-minute or high-emotion pitch variations), extracts a 60-second clip from the source video, crops it into vertical 9:16, and overlays viral captions.

8. Faceless Video / AI Video Generator

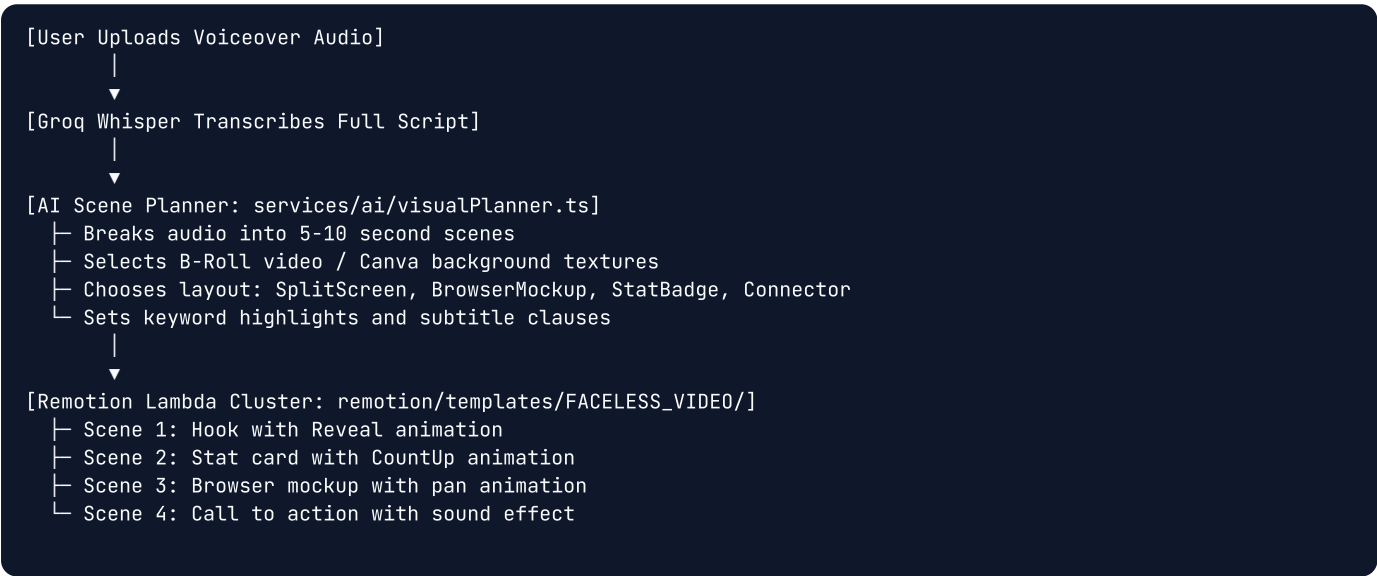
16:9 LANDSCAPE

FULL AI SCENE DIRECTOR

8.1 Identity & Specs

- **Dashboard Mode:** mode: "facelessVideo" / mode: "aiVideoGenerator"
- **Remotion Composition ID:** FACELESS-VIDEO
- **Aspect Ratio:** 1920 × 1080 (16:9 Landscape)
- **Duration:** Up to **20 Minutes**
- **Inputs:** Voiceover Audio (MP3/WAV) + Topic Script.

8.2 Architecture & Scene Planning Engine



8.3 Common Errors & Technical Fixes

Error: Lambda Memory Crash on 15+ Min Faceless Videos
Root Cause: Pre-loading hundreds of high-resolution B-roll images and Canva backgrounds causes Chromium to exceed 2048 MB RAM.
Fix: Asset sequence rendering uses lazy-loaded image components with Remotion's `Img` component, and AWS Lambda function memory is allocated at **3008 MB**.

9. Image To Video AI

16:9 LANDSCAPE

MOTION PHOTO STORY

9.1 Identity & Specs

- **Dashboard Mode:** mode: "imageToVideoAi"
- **Remotion Composition ID:** IMAGE-T0-VIDEO-AI
- **Inputs:** 3 to 10 Images + Narration Audio Voiceover.

- **Effect:** Smooth Ken Burns pan/zoom camera motion, cross-dissolve scene transitions, and synchronized kinetic captions.

10. Creator Background Replace

9:16 VERTICAL

STUDIO VFX

10.1 Identity & Specs

- **Dashboard Mode:** mode: "creatorBackgroundReplace"
 - **Inputs:** 1 Creator talking video + 1 Background image or video.
 - **Technology:** AI video segmentation removes the messy home background and places the creator inside a luxury studio, modern office, or podcast room.
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11. AI Audio Cleaner

AUDIO PROCESSOR

STUDIO GRADE

11.1 Identity & Specs

- **Dashboard Mode:** mode: "audioClean"
- **Inputs:** Raw voiceover or podcast audio (up to 15 mins).
- **What it does:**
 - **De-Noising:** Eliminates AC hum, computer fan noise, and background street rumble.
 - **De-Reverb:** Removes room echo from untreated rooms.
 - **Dead Pause Cutting:** Automatically detects silent gaps > 0.6s and trims them for snappy podcast pacing.

Master Engineering Comparison Matrix

Template Name	Composition ID	Media Required	Skip AI Planner?	Fast Path?	AWS Concurrency
YouTube Subtitle	AUTO-CAPTION-GENERATOR-LANDSCAPE	Video (16:9)	Yes	Yes (Groq + Lambda)	4 – 10
Auto Caption	AUTO-CAPTION-GENERATOR	Video (9:16)	Yes	Yes	2 – 4
Compare Explainer	comparisonImages	Audio + 2 Images	No	No	2
Long Video Promo	LONG-VIDEO-PROMO	Video + Thumbnail	Yes	Yes (Instant)	2
Whiteboard	WHITEBOARD-VIDEO	Audio / Video	Yes	No	2
Typography	TYPOGRAPHY-VIDEO	Audio / Video	Yes	No	2
Long Video Clips	LONG-VIDEO-CLIPS	Long Video	Yes	No	4
Faceless Video	FACELESS-VIDEO	Audio + Script	No (Deep LLM)	No	6 – 10
Image To Video AI	IMAGE-T0-VIDEO-AI	Audio + Images	No	No	4
Background Replace	Custom Layer Stack	Video + Background	Yes	No	2